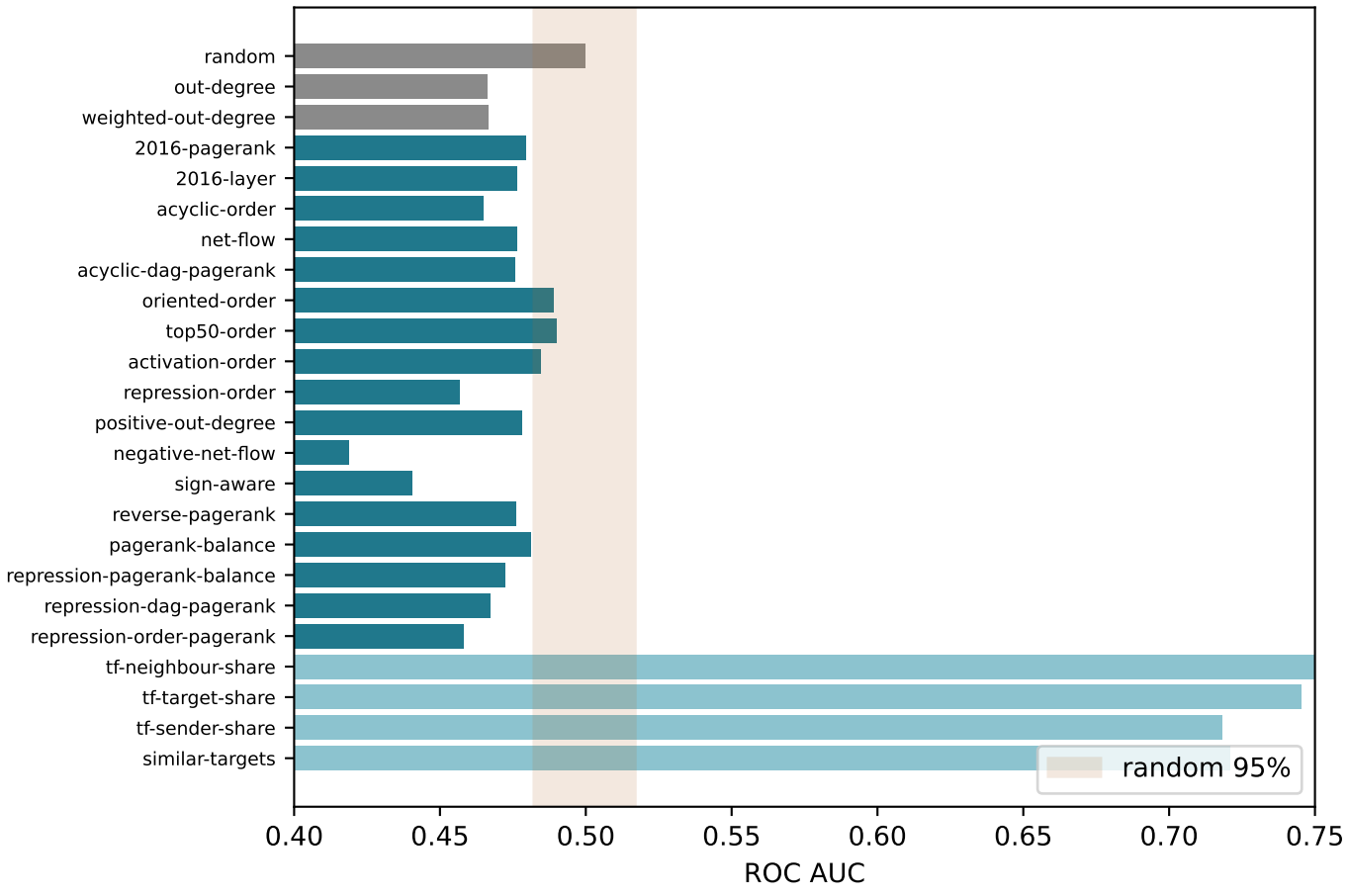
Regulator ranking against reference 'lambert-2018'

2026-11-10 · 11,949 genes ranked, 1,146 in lambert-2018-tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	9.5 (4–16)	0.500 (0.482–0.517)				
out-degree	4	0.466	0.989	1.000	-0.000 (-0.001, +0.001)	0.728
weighted-out-degree	4	0.466	0.989	1.000	+0.000 (+0.000, +0.000)	
2016-pagerank	11	0.479	0.357	0.985	+0.013 (-0.008, +0.033)	0.268
2016-layer	9	0.476	0.626	0.994	+0.010 (-0.009, +0.030)	0.324
acyclic-order	2	0.465	1.000	1.000	-0.002 (-0.012, +0.008)	0.736
net-flow	3	0.476	0.998	0.994	+0.010 (-0.007, +0.025)	0.286
acyclic-dag-pagerank	4	0.476	0.989	0.995	+0.009 (+0.000, +0.017)	0.042
oriented-order	6	0.489	0.930	0.887	+0.023 (+0.001, +0.042)	0.050
top50-order	11	0.490	0.357	0.864	+0.023 (+0.003, +0.042)	0.024
activation-order	12	0.485	0.231	0.956	+0.018 (+0.005, +0.030)	0.006
repression-order	4	0.457	0.989	1.000	-0.010 (-0.027, +0.007)	0.270
positive-out-degree †	7	0.478	0.857	0.990	+0.012 (+0.008, +0.015)	< 0.001
negative-net-flow †	3	0.419	0.998	1.000	-0.048 (-0.065, -0.032)	< 0.001
sign-aware †	3	0.441	0.998	1.000	-0.026 (-0.034, -0.018)	< 0.001
reverse-pagerank †	5	0.476	0.973	0.994	+0.010 (+0.002, +0.017)	0.012
pagerank-balance †	3	0.481	0.998	0.981	+0.015 (-0.005, +0.033)	0.142
repression-pagerank-balance †	11	0.472	0.357	1.000	+0.006 (-0.013, +0.024)	0.548
repression-dag-pagerank †	8	0.467	0.752	1.000	+0.001 (-0.017, +0.018)	0.970
repression-order-pagerank †	4	0.458	0.989	1.000	-0.008 (-0.025, +0.008)	0.330
tf-neighbour-share ‡	95	0.750	< 0.001	< 0.001	+0.283 (+0.260, +0.306)	< 0.001
tf-target-share ‡	85	0.745	< 0.001	< 0.001	+0.279 (+0.254, +0.301)	< 0.001
tf-sender-share ‡	88	0.718	< 0.001	< 0.001	+0.252 (+0.230, +0.274)	< 0.001
similar-targets ‡	94	0.721	< 0.001	< 0.001	+0.254 (+0.230, +0.280)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

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Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	31,683	0.036 (3.49×)	4.09×	0.69×	0.85 (0.83)
resolve_reciprocals → drop_sinks → prune_triangles	30,471	0.002 (1.11×)	3.06×	0.69×	0.65 (0.49)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	21,539	0.001 (0.93×)	3.04×	0.68×	0.57 (0.46)
absolute	31,683	0.036 (3.49×)			
absolute → minimum_feedback_order	30,654	0.025 (3.11×)			
orient_reciprocals ratio=2.0 → absolute	30,646	0.014 (2.11×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	30,646	0.014 (2.12×)			
top_outgoing k=50 → absolute	31,683	0.017 (5.02×)			
top_outgoing k=50 → absolute → minimum_feedback_order	30,646	0.011 (4.34×)			
keep_sign sign=1 → absolute	31,625	0.035 (4.08×)			
keep_sign sign=1 → absolute → minimum_feedback_order	30,211	0.024 (3.64×)			
keep_sign sign=-1 → absolute	28,295	0.001 (0.71×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	28,005	0.001 (0.74×)			
keep_sign sign=-1	28,295	0.001 (0.71×)		0.71×	0.43 (0.43)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 1,588,586 edges (316,508 negative), largest strong component 11,945.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 1,588,586 edges (316,508 negative), largest strong component 11,945.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 341,151 edges (291,253 negative), largest strong component 11,653.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 341,151 edges (291,253 negative), largest strong component 11,653; DAG keeps 196,254 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,588,586 edges (0 negative), largest strong component 11,945; DAG keeps 1,205,421 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 1,588,586 edges (0 negative), largest strong component 11,945.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 1,588,586 edges (0 negative), largest strong component 11,945; DAG keeps 1,205,421 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,014,657 edges (0 negative), largest strong component 6,231; DAG keeps 997,725 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 476,796 edges (0 negative), largest strong component 11,834; DAG keeps 352,701 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,272,078 edges (0 negative), largest strong component 11,851; DAG keeps 937,305 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 316,508 edges (0 negative), largest strong component 9,722; DAG keeps 282,332 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 1,588,586 edges (316,508 negative), largest strong component 11,945.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 1,588,586 edges (316,508 negative), largest strong component 11,945.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 1,588,586 edges (316,508 negative), largest strong component 11,945.

reverse-pagerank: Highest reverse PageRank (raw graph)